

Input:2

```
real    0m0.017s
user    0m0.010s
sys     0m0.006s
```

Input:10

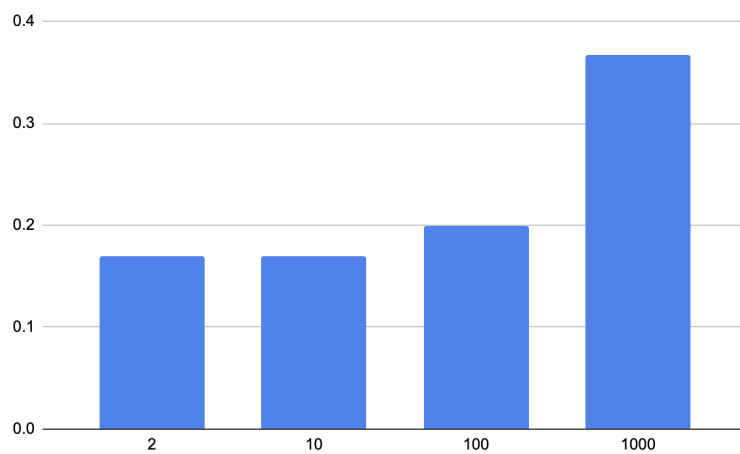
```
real    0m0.017s
user    0m0.011s
sys     0m0.005s
```

Input:100

```
real    0m0.020s
user    0m0.010s
sys     0m0.009s
```

input:1000

```
real    0m0.367s
user    0m0.257s
sys     0m0.100s
```



These results meet my expectations in the fact that this program is efficient and scalable. Even after multiplying the input size by 500, the output time is no more than 2x slower. This holds up to the asymptotic complexity of $O(n^2)$